

Generation Interconnection System Impact Study Report

For Queue Project Q15-0439

Airline 230 kV

May 03, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0439. The Interconnection Customer proposes a 128.3 MW solar photovoltaic facility interconnecting at the Airline 230 kV substation in the Ohio area. The request was received on February 18, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 250 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,272,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MIDDLE FORK-FISHKILL PLAINS 345kV line	SIMPSON STUAR TAP-FISHKILL PLAINS 345kV	112.0	1,241	6.1
FISHKILL PLAINS-DILLARD-ORANGE 345kV	INTERNATIONAL PAPER-FLATLICK 500kV L	105.5	2,489	6.2
GREENLY-DRESDEN 500kV	DRESDEN-INTERNATIONAL PAPER 500kV Line	104.9	2,175	31.0
OMNI-HACKBERRY-GREENLY 500kV	INTERNATIONAL PAPER-DRESDEN 500kV	101.9	2,579	29.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FISHKILL PLAINS-MIDDLE FORK 345kV L	Rebuild line with higher-capacity conductor	\$41,858,000	1% (\$600,000)
FISHKILL PLAINS-DILLARD-ORANGE 345kV	Reconductor line	\$36,489,000	1% (\$319,000)
GREENLY-DRESDEN 500kV	Reconductor line	\$29,925,000	2% (\$748,000)
OMNI-HACKBERRY-GREENLY 500kV line	Reconductor line	\$24,711,000	2% (\$605,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$2,272,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.